The Timestamp Is Not the Event

Uncertainty-aware temporal fusion for asynchronous industrial sensors at the physical edge

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Thesis. Multimodal industrial fusion should treat event time as an interval or distribution with provenance, not as a single perfectly known scalar. Clock synchronization improves comparability; it does not eliminate acquisition latency, buffering, timestamp origin, clock-domain, or retrospective-entry uncertainty.

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Abstract

Industrial edge systems increasingly combine vibration, acoustic, thermal, image, distance, environmental, network, and maintenance data. These streams rarely share the same sampling cadence, transport path, buffering behavior, or timestamp origin. Yet many fusion pipelines collapse each observation to a single timestamp and then apply a fixed coincidence window. This paper argues that such a scalar-time model can create false physical correlations even when every software component is functioning correctly. The proposed alternative is **uncertainty-aware event time**: each observation carries a best estimate of physical event time together with a bounded uncertainty model, clock domain, timestamp origin, and transformation history. Fusion then asks whether two observations are temporally compatible at a stated confidence, rather than whether their scalar timestamps fall inside an arbitrary window. The paper relates this proposal to IEEE 1588 Precision Time Protocol, NIST time-traceability work, OPC UA source timestamps, ROS approximate-time synchronization, and recent research on asynchronous sensor fusion under unknown temporal misalignment. It develops an architecture, compatibility rule, implementation path, limitations, and evaluation methodology for heterogeneous industrial edge sensing. SynapticSteel TelePod is used as a concrete design context; the uncertainty-aware fusion layer described here is a proposed architecture, not a claim of completed production implementation.

Evidence boundary. The cited standards and technical work establish that clock synchronization, timestamp provenance, approximate synchronization, and temporal misalignment are real engineering concerns. The specific interval/distribution representation and fusion policy proposed here are an architectural synthesis and should be evaluated experimentally before stronger performance claims are made.

1. The physical event and the recorded timestamp are different objects

A sensor reading participates in at least three times: the time the physical phenomenon occurred, the time the sensing subsystem sampled or integrated it, and the time software received or persisted the resulting message. For a camera there may also be exposure start, exposure midpoint, frame completion, encoder output, USB arrival, decode completion, and application callback. For a BLE device, radio scheduling and host buffering intervene. For a maintenance note, the recorded time may be entered minutes after the wrench was turned.

The engineering mistake is not using timestamps. The mistake is allowing one timestamp field to silently stand in for all of these meanings. OPC UA explicitly distinguishes source timestamps from server timestamps and specifies that the source timestamp should be generated as close as possible to the data source and propagated unchanged through another OPC UA server [1]. That distinction is evidence that timestamp origin is semantically important, not merely an implementation detail.

IEEE 1588-2019 addresses a different layer: synchronization of clocks in packet-based measurement and control systems, supporting sub-microsecond synchronization in suitable deployments [2]. NIST similarly emphasizes that distributed measurement becomes easier when components share a common sense of time [3]. But synchronized clocks do not make all timestamps physically equivalent. A perfectly synchronized host clock cannot reveal when a buffered sensor actually integrated the quantity unless the acquisition path exposes that information.

Time field	Meaning	Typical uncertainty source
t_event	Physical phenomenon of interest	Often latent; may span an interval
t_acquire	Sensor sampling/integration time	Exposure/integration window, device oscillator
t_source	Timestamp applied near data source	Clock error, driver semantics

Time field	Meaning	Typical uncertainty source
t_receive	Host/application receipt	Transport, queues, USB/BLE/network jitter
t_persist	Storage/event-log commit	Scheduler and storage latency
t_human	Human-entered event time	Recall and entry delay

Table 1. A single observation can legitimately have several time coordinates. Collapsing them destroys provenance.

2. Why approximate synchronization is necessary - and insufficient

Robotics middleware provides a practical example of the problem. ROS 2's `ApproximateTimeSynchronizer` matches messages using timestamps and a configurable tolerance; its documentation also warns against substituting arrival time for message time because delays are unpredictable [4]. This is a useful engineering mechanism when the application accepts a bounded slop window.

However, a fixed slop parameter answers only: *are these timestamps numerically close enough?* It does not answer: *how certain are we that the underlying physical events overlapped?* Two messages can have identical host timestamps yet originate from different acquisition intervals. Conversely, two measurements whose recorded timestamps differ by 80 ms might represent overlapping physical integration windows if one path is known to add 100 ms of deterministic buffering.

Recent signal-processing research makes the issue explicit. A 2026 IEEE Transactions on Signal Processing paper treats asynchronous sensor fusion when temporal misalignment is not precisely known and jointly estimates the time offset during fusion [5]. Although that work targets tracking rather than industrial condition monitoring, it demonstrates that unknown temporal offset can be a first-class estimation variable rather than a nuisance hidden inside a fixed window.

3. Proposed representation: event time as evidence

For heterogeneous industrial sensing, the minimal record should preserve a best estimate and an uncertainty description rather than only `timestamp`. A lightweight implementation need not carry a full probability density for every sample. It can begin with a bounded interval and provenance fields.

Field	Example	Purpose
event_time_estimate	2026-09-18T14:03:22.417Z	Best estimate of physical event time
uncertainty_before	12 ms	How far earlier the event may plausibly be
uncertainty_after	35 ms	How far later it may plausibly be
timestamp_origin	device_source / host_receive / human	Semantic origin
clock_domain	PTP-A / host-monotonic / BLE-device	Identify clock relationship
sync_state	locked / holdover / unsynchronized /unknown	Describe clock confidence
acquisition_window	10 ms exposure / 2 s thermal integration	Represent temporal support
transform_lineage	driver -> resampler -> feature	Preserve time-affecting transformations

Table 2. Proposed minimum temporal-evidence fields. Exact names are illustrative, not a standard.

A bounded observation can be represented as I = [t - u-, t + u+]. For two observations A and B, a simple temporal-compatibility test asks whether their intervals overlap. A stronger implementation

can use probability distributions and compute the probability that their latent event times differ by less than a physically meaningful tolerance.

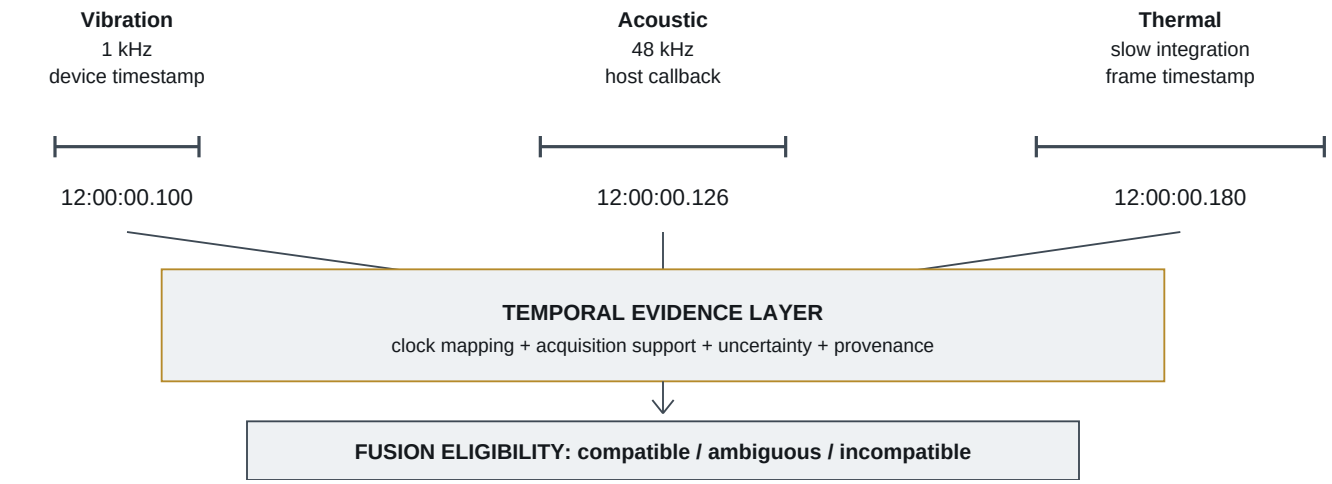


Figure 1. Fusion occurs after temporal evidence is normalized. The visual interval lengths represent uncertainty/support, not sampling rate.

4. Synchronization is a measurement with an uncertainty budget

A system can improve its temporal evidence by synchronizing clocks, but synchronization quality should itself be represented as evidence. IEEE 1588 supports precise clock synchronization and includes mechanisms for heterogeneous clocks and path correction [2]. NIST's traceability work states a stricter metrological principle: traceability requires an unbroken calibration chain with documented measurement uncertainty [6]. The industrial edge does not need national-metrology rigor for every sensor, but the conceptual lesson is valuable: a claim of 'synchronized' should have an error budget.

For a practical TelePod-class device, that budget might include device-clock offset, oscillator drift since last synchronization, bus latency bounds, driver timestamp placement, acquisition-window duration, and any known buffering. If a device cannot expose source time, the system should say so. Unknown is a legitimate state; false precision is not.

5. Competing approaches

One host clock for everything. Timestamping all data on receipt is simple and sometimes adequate on tightly integrated hardware. It becomes misleading when paths have materially different or variable latency.

PTP everywhere. Precision Time Protocol is powerful where hardware and network support exist, but it cannot retroactively expose hidden sensor integration time or human-event uncertainty. It solves clock alignment, not every temporal semantic.

Fixed coincidence windows. Easy to implement and efficient. They work when timing error is known and stationary. They become brittle across sensors with different acquisition windows and changing transport conditions.

Learn the offset. Joint offset estimation, as in recent asynchronous-fusion research [5], can be superior when sufficient correlated dynamics exist. It adds computation and can fail when the streams have weak shared structure.

Resample everything to a common rate. Useful for downstream models, but interpolation can manufacture apparent simultaneity. Resampling should occur after provenance is preserved, not instead of preserving it.

6. Causal restraint: temporal compatibility is not causality

Even perfect temporal alignment does not establish that one signal caused another. A motor-current transient, vibration spike, acoustic impulse, and camera motion may all be consequences of a common process transition. Temporal compatibility should therefore be treated as an eligibility condition for fusion or causal analysis, not as causal proof.

This distinction is particularly important in industrial troubleshooting, where attractive stories form quickly. A multimodal system can strengthen evidence by showing that independent channels changed within compatible physical-time intervals. It should still preserve competing explanations and avoid promoting correlation into causal authority without intervention or domain evidence.

7. Relevance to SynapticSteel TelePod

TelePod is a natural architectural context because its sensor work spans high-rate vibration/IMU, audio, camera, time-of-flight, thermal, environmental, and machine-state observations. Those modalities differ sharply in cadence and transport. The system's design emphasis on persistent temporal behavior makes timestamp semantics unusually consequential.

The maturity boundary is explicit. **Implemented or prototyped work** includes multiple physical sensor collectors, local state/services, and experiments connecting modalities to TelePod-class edge compute. **Experimental work** includes multimodal fusion, temporal models, and operator visualizations. **Proposed by STP-003** is a unified temporal-evidence layer that carries uncertainty bounds, timestamp origin, clock domain, synchronization state, and acquisition support into fusion eligibility. It should not be described publicly as an implemented TelePod capability until it is built and validated.

The proposal also fits the architectural separation already used elsewhere: collectors observe; a temporal-evidence layer characterizes when observations plausibly occurred; fusion combines compatible evidence; models interpret; authorities govern stronger operational claims. No model becomes the time authority merely because it can fit asynchronous data.

8. Evaluation methodology

Experiment	Measurement
Known-delay injection	Add controlled transport delays/jitter and measure recovered event-time compatibility
Cross-modal impulse	Apply a physical impulse visible to vibration, audio, camera/ToF; measure alignment error
Clock degradation	Disable synchronization or enter holdover; verify uncertainty expands rather than silentlystaying fixed
Buffered camera path	Introduce deterministic and variable frame delay; compare receipt-time vs source-time fusion
Retrospective maintenance marker	Enter approximate intervention time; verify causal confidence remains bounded
Ablation	Compare fixed-window fusion against uncertainty-aware eligibility for false matches andmissed matches

Table 3. A controlled bench evaluation can test the architecture without claiming field-scale prognostic performance.

Primary metrics should include false temporal-match rate, missed-match rate, calibration of claimed compatibility probability, sensitivity to clock degradation, compute/memory overhead, and downstream change in false multimodal anomaly correlation. Ground-truth physical triggers should be generated by hardware or instrumented actions rather than inferred from the same timestamps being evaluated.

9. Limitations and counterarguments

Uncertainty models can be wrong. A precise-looking interval is harmful if its bounds are guessed. Bounds need provenance and conservative defaults.

Complexity has a cost. Many industrial applications do not need microsecond-level alignment. The proposal is most valuable where heterogeneous paths and cross-modal reasoning make timing ambiguity material.

Independence assumptions may fail. Timing errors can be correlated because streams share a USB hub, scheduler, clock source, or network path. Probabilistic implementations must not blindly multiply correlated confidences.

Hardware support varies. Some low-cost sensors expose no source timestamp. The architecture must support degraded temporal evidence instead of requiring ideal hardware.

Latency and time error are different. A system may tolerate high latency if event time is trustworthy; conversely, a low-latency stream may have poor event-time semantics. Operational dashboards should not conflate the two.

Storage overhead is real. Per-sample uncertainty metadata may be excessive for high-rate channels. Efficient implementations can attach timing contracts to stream segments or epochs and record exceptions per sample.

10. Conclusion

Industrial multimodal intelligence depends on knowing not only what was measured, but when the underlying physical event plausibly occurred. A scalar timestamp often hides acquisition windows, buffering, clock domains, transport jitter, retrospective entry, and synchronization quality. Precise clocks reduce one part of the problem; they do not erase the rest.

The proposed temporal-evidence layer makes uncertainty and provenance explicit before fusion. It allows a system to say that two observations are compatible, ambiguous, or incompatible in physical time, and to lower confidence when timing evidence degrades. This is a modest architectural change with an important epistemic effect: the system becomes less likely to manufacture simultaneity from software timestamps.

For persistent physical intelligence, time should be treated like any other measurement: useful, calibratable, provenance-bearing, and never more certain than the evidence permits.

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Related SynapticSteel Technical Papers

STP-001 - Failure Metabolism. Develops evidence-preserving operational memory and causal lineage. STP-003 narrows that concern to the temporal provenance required before cross-modal evidence is allowed to reinforce a claim.

STP-002 - Baseline Is Not Health. Separates stable observation from health assertion and argues that timing uncertainty matters around interventions. STP-003 develops the timing problem directly: the event time itself may be uncertain even when the stored timestamp is exact.

Publication note

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